

Smart Agriculture – Automatic Crop, Fertilizer, and Pesticide Recommendation Using Machine Learning Techniques

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By

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Abstract

India, as an agricultural nation, has its economy largely reliant on the production of agricultural yields and agro-industry products. Every farmer is keen to determine the most suitable crop for their land. In the field of agriculture, crop losses often occur due to the incorrect selection of crops for a specific plot of land. Farmers typically lack knowledge about the specific requirements of crops, including minerals, soil moisture, and other necessary soil conditions.

This lack of awareness can lead to both mental and financial distress for farmers. Another significant challenge faced by farmers is the potential impact of pests and diseases on their crops, often going unnoticed in the early stages.

Machine learning algorithms play a crucial role in analyzing diverse sets of data, encompassing soil types, and weather patterns, to make precise recommendations regarding which crop is best suitable to grow. This capability empowers farmers to optimize their farming practices, ultimately leading to increased yield, cost reduction, and a reduced environmental footprint. Furthermore, machine learning algorithms extend their utility beyond yield prediction by offering recommendations for an ideal crop rotation schedule, fertilizer type and quantity, and irrigation timetable tailored to specific regions. This is accomplished through a thorough analysis of soil composition, nutrient levels, and weather patterns, enabling farmers to boost their yield while minimizing costs and lessening environmental impacts. In the realm of pest and disease management, machine learning proves invaluable by swiftly detecting signs of pests and diseases based on crop image analysis and subsequently suggesting appropriate treatment measures. This empowers farmers to proactively address issues, reducing crop damage and bolstering overall yield.

This analysis encompasses a range of related attributes, including location and the pH value, which determines soil alkalinity. Additionally, it involves assessing the percentages of key nutrients such as Nitrogen (N), Phosphorus (P), and Potassium (K). Location data is supplemented with third-party applications, such as APIs for weather and temperature information, soil type, nutrient composition of the soil in the region, regional rainfall patterns, and soil composition. All these data attributes are subject to thorough analysis.

To create an accurate model, the dataset is trained using various appropriate machine learning algorithms such as Support Vector Machines (SVM), Random Forest, k-Nearest Neighbors (KNN), and a voting classifier. The resulting system provides precise and reliable predictions regarding crop yields and offers valuable recommendations on the required fertilizer ratios based on atmospheric and soil parameters specific to the land. This, in turn, leads to increased crop yields and higher revenues for farmers.

In conclusion, machine learning techniques are pivotal in steering the agriculture industry towards a sustainable future. They achieve this by fine-tuning farming practices, elevating crop yields, and mitigating environmental impact. In a world confronted with increasingly complex challenges, these techniques assume greater significance than ever before.

In summary, this proposed system takes into account soil quality data, including Nitrogen, Phosphorus, Potassium, and pH values, as well as weather-related information, such as rainfall, temperature, and humidity, to make informed predictions about the most suitable crops for cultivation. This system will also give some fertilizer-based suggestions, depending upon the NPK values of the soil and crop to be grown. Further, this system will also predict the pesticide that can be used, based upon the pest identified on the crop, by taking its image as an input.

Keywords: SVM, KNN, Data mining, Machine Learning, Random Forest, Voting Classifier, APIs

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List of Abbreviations

1. SVM- Support Vector Machine
2. KNN- K-Nearest Neighbour
3. NPK- Nitrogen-Phosphorous-Potassium
4. GDP- Gross Domestic Product
5. EDA- Exploratory Data Analysis
6. CART- Classification and Regression Trees
7. UI- User Interface
8. CNN- Convolutional Neural Network
9. ISO- International Organization for Standardization

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Chapter 1

Introduction

1.1 Overview of Work

Agriculture holds immense importance in India due to its pivotal role in the country's economy, livelihoods, and food security. Agriculture is one of the largest sectors in the Indian economy, contributing significantly to the Gross Domestic Product (GDP). It provides employment to a substantial portion of the population, particularly in rural areas. A large percentage of India's population depends directly or indirectly on agriculture for their livelihood. Small and marginal farmers, along with agricultural laborers, constitute a significant part of the workforce. Agriculture supplies raw materials for various industries, including textiles, agro-processing, and food manufacturing. This linkage between agriculture and industry is essential for economic growth.

India exports a variety of agricultural products, such as rice, wheat, spices, and tea. These exports contribute to foreign exchange earnings, bolstering the nation's foreign reserves. With climate change becoming a global concern, agriculture in India is adapting to climate-resilient practices and sustainable farming methods. This is crucial to ensure food production in the face of changing weather patterns. The agriculture sector faces global challenges like feeding a growing population and ensuring food safety. Sustainable agricultural practices are essential to meet these challenges and contribute to global food security.

Thus, agriculture in India is not just an economic activity; it is a way of life and an integral part of the country's identity. Its significance extends beyond food production and encompasses economic development, social stability, and environmental sustainability. To address the evolving needs and challenges, India continues to focus on modernizing its agricultural sector and promoting sustainable practices.

In India, the significance of agriculture can be understood by the ever-expanding population and the heightened demand for food. Nevertheless, crop yields face adverse effects stemming from diseases induced by bacteria, viruses, and fungi. To address this issue, the adoption of machine learning technology for plant disease detection can play a pivotal role in disease identification, ultimately leading to increased crop yields. Additionally, the recent fluctuations in crop prices and the unpredictability of climate patterns have further compounded the challenges faced by farmers, necessitating the development of a system to aid farmers in achieving high-yield production and safeguarding their crops.

Farmers grapple with crop losses caused by uncertainties, limited knowledge concerning plant diseases, and unpredictable yields. India's economy heavily relies on agriculture, yet traditional methods for crop management, such as fertilizer usage and crop selection, continue to be prevalent among farmers.

The consequences of inadequate plant management and suboptimal fertilizer application have manifested as drastic fluctuations in crop prices and an increase in crop damage in recent years. The overarching purpose of the forecasting system is to provide farmers with insights into yield expectations and strategies for risk management.

Leveraging machine learning algorithms to scrutinize extensive datasets and furnish accurate predictions regarding crop yields assumes paramount importance in research endeavors dedicated to upholding food security. Precision agriculture, rooted in information technology techniques, strives to augment crop yields. Our system prioritizes the economic viability and sustainability of farmers and offers geographically specific crop forecasts, accompanied by graphical data analysis.

The fertilizer recommendation system harnesses modern information technology, as well as soil quality assessment, and fertilizer theory to enhance soil nutrient management. It is meticulously crafted as a fertilization model, underpinned by expert knowledge and practices rooted in agronomy research. Plant infections and diseases can originate from a spectrum of sources, including living organisms and non-living factors such as soil pH, nutrient deficiencies, and prevailing weather conditions.

Thus, the proposed system under consideration utilizes input data pertaining to soil quality and weather-related information. This encompasses various aspects of soil quality, including Nitrogen, Phosphorus, Potassium, and pH levels. The weather-related data incorporates details such as rainfall, temperature, and humidity. Further, my model will also recommend some suggestions to the farmers regarding the use of fertilizer based on N, P, K values of soil and type of crop. It will also recognize the pests on crops and will recommend particular pesticides to prevent any losses from the pests.

It is important to note that for our project, we source datasets from the Kaggle website.

1.2 Motivation of the work

Crop prediction is a prevalent issue that demands attention. Farmers are naturally curious about the expected yield during the growing season. In the past, yield predictions relied heavily on a farmer's long-term experience with specific crops and local climatic conditions. This approach led farmers to focus on yield predictions rather than broader crop predictions through the existing system. However, without accurate crop predictions, it is challenging to optimize yield. Furthermore, the existing systems often do not account for crucial factors like pesticides, environmental conditions, and meteorological parameters relevant to crop cultivation.

To promote and enhance agricultural production at a faster rate, addressing this issue becomes crucial. The success of any crop's production is influenced by factors such as domain-specific interest and yield enhancement. In India, expanding the area under any particular crop is only possible through either increasing cropping intensity or replacing one crop with another. Consequently, fluctuations in crop productivity persist and create significant challenges for the region, leading to considerable distress.

The accurate prediction of which crops are most suitable for a specific region based on NPK values and weather conditions can significantly enhance agricultural productivity. It helps farmers make informed decisions about what to plant, which, in turn, leads to higher yields and profits.

By recommending crops based on soil nutrient levels and expected weather patterns, the system can assist farmers in using resources such as fertilizers and irrigation more efficiently. This minimizes resource wastage and reduces production costs. Precise crop recommendations promote sustainable farming practices. When farmers cultivate crops suitable for their land and climate, it reduces the need for excessive

chemical inputs and minimizes environmental impact.

Weather patterns are becoming increasingly unpredictable due to climate change. A crop recommendation system can help farmers mitigate risks associated with extreme weather events by suggesting crops that are more resilient to adverse conditions.

The use of machine learning techniques can lead to more accurate predictions, resulting in higher crop yields. This not only benefits farmers by increasing their income but also contributes to food security. Machine learning systems can be designed to be user-friendly, making it easier for farmers, including those with limited technical expertise, to access and implement crop recommendations. Incorporating NPK values and weather data into crop recommendations aligns with a more scientific approach to agriculture. This is especially important in modern farming, where precision and efficiency are key.

Hence, there is a pressing need to adopt effective techniques for crop prediction to overcome the existing challenges and uncertainties in agriculture.

1.3 Literature Review

The proposed approach in the paper [1] aims to address the challenges faced by Indian farmers in making informed decisions about which crops to grow during different seasons. The proposed system integrates two sub-systems to provide crop recommendations: the Crop Suitability Predictor and the Rainfall Predictor.

The authors use the 'India Agriculture and Climate Data Set' which contains historical records of soil and meteorological parameters from a thirty-year period. These parameters include soil type, aquifer thickness, soil pH, thickness of topsoil, precipitation, temperature, and location parameters.

To perform crop recommendations, the authors use machine learning algorithms that support multi-label classification. The algorithms considered are Decision Tree, K Nearest Neighbor (K-NN), Random Forest, and Neural Network. After applying the data to different machine learning algorithms, a trained model for crop recommendation is obtained. Farmers can input their farm's soil type, aquifer characteristics, topsoil thickness, pH, and the predicted rainfall. The trained model then provides crop recommendations based on these inputs.

The Rainfall Predictor sub-system uses a training dataset comprising 117 years of month-wise rainfall data for all 29 states in India, obtained from a meteorological dataset. To predict rainfall, a linear regression algorithm is used. The algorithm establishes a linear relation between Indian states and the monthly precipitation values. After fitting the training dataset to the linear regression algorithm, a trained rainfall predictor model is obtained. When an Indian state is input into the model, it provides 12 float values, each corresponding to the predicted rainfall (in mm) for one of the twelve months in that state.

The Rainfall Predictor achieved an accuracy of 71%, while the Crop Suitability Predictor was implemented using the Neural Network algorithm due to its highest accuracy (91%) among the algorithms tested.

The research paper [2] focuses on addressing the challenges faced by farmers in India due to factors like erroneous crop selection, lack of awareness about soil requirements, and the threat of pests and diseases affecting crops. The authors propose a recommendation system that helps farmers choose suitable crops for their land and provides guidance on pest control techniques. The paper also evaluates the effectiveness of different machine learning algorithms in crop prediction.

The introduction of the paper highlights the significance of agriculture in India's economy, with a substantial contribution to GDP and employment. It emphasizes the need to tackle issues such as crop loss, soil exhaustion, and pest infestations that lead to distress among farmers. The Food and Agriculture Organization (FAO) data is cited, revealing significant food wastage globally due to these challenges. The authors stress that addressing these issues is vital, and their paper proposes a solution through a recommendation system.

The dataset used for crop prediction includes essential soil attributes like soil color, pH, average rainfall, and temperature. These attributes are crucial for determining suitable crops for cultivation. The paper also describes the classification algorithms employed for crop prediction: Support Vector Machine (SVM), Decision Tree, and Logistic Regression.

The authors have proposed their work, which involves training the model with a dataset and applying SVM and Decision Tree algorithms for crop prediction. The authors found that the SVM algorithm provided more accurate results. After training the model with the dataset, it is applied to make predictions based on the soil parameters provided. The model takes soil parameters as input, stored in a .csv file, to predict the most suitable crop for cultivation. It also provides information about common pests affecting the predicted crop and suggests pest control methods.

The Result Analysis highlights that the model achieved an accuracy rate of 89.66%. SVM performed the best among the algorithms tested.

In conclusion, the paper emphasizes the importance of helping farmers make informed decisions in crop selection and pest control. The proposed recommendation system shows promise in addressing these challenges. Future work is expected to focus on expanding the model with additional soil attributes and larger datasets.

The research paper [3] explores the development of an integrated agriculture system designed to address the challenges faced by farmers in India. Agriculture holds critical importance in the country due to a growing population and increasing food demand. However, this sector is plagued by issues such as crop diseases caused by bacteria, viruses, and fungi, as well as price fluctuations and uncertain climate trends. In response to these challenges, the paper proposes leveraging machine learning techniques to empower farmers with the tools needed to make informed decisions about crop selection, fertilizer usage, and plant disease management, ultimately leading to higher crop yields and contributing to food security.

One of the central challenges for farmers is the lack of access to accurate information and resources. This paper emphasizes that farmers often struggle with limited knowledge of plant diseases, uncertainties in crop yields, and unpredictable price fluctuations. The proposed system aims to address these issues by providing accurate models and recommendations based on national air and soil parameters, offering a data-driven approach to crop management and fertilization. The integration of machine learning algorithms plays a

pivotal role in achieving these goals.

The literature review within the paper highlights the importance of machine learning in agriculture and presents a range of studies and systems that have focused on crop recommendation, disease detection, and fertilizer optimization. These existing research efforts serve as valuable references for building an improved system capable of addressing the specific challenges faced by Indian farmers.

The core of the paper lies in the description of the proposed system, which is structured into three main components: crop recommendation, fertilizer recommendation, and plant disease detection. The crop recommendation system allows users to input soil data, and based on this data, the system provides recommendations for the most suitable crops to cultivate. The fertilizer recommendation system leverages soil quality assessment and fertilization theory to predict nutrient deficiencies or excesses and recommend appropriate improvements. Additionally, the plant disease detection system empowers farmers to identify diseases in their crops by submitting images of affected plant leaves.

The paper also provides insights into the algorithms used and their respective accuracy rates. Notably, the LightGBM algorithm is employed for crop prediction, achieving an accuracy rate of 94.86%. The Random Forest Classifier is used for fertilizer prediction, with an impressive accuracy of 96%. For plant disease detection, the AlexNet classifier is utilized, delivering an accuracy rate of 92.2%.

The system is supported by a well-structured database schema to efficiently manage and store the data generated by the various components. The authors also discuss the system's practical application and how it is accessible to farmers through a user-friendly interface.

In conclusion, the research paper presents a comprehensive solution to the challenges faced by Indian farmers in the agriculture sector. By harnessing the power of machine learning and data-driven recommendations, the system offers valuable tools for crop selection, fertilizer management, and disease detection. This has the potential to significantly impact agriculture in India, enhance food security, and improve farmers' livelihoods. Furthermore, the authors suggest future enhancements, including the integration of IoT sensors to provide real-time data and further improve system accuracy.

In this paper [4], the use of pesticides and chemical treatments to control pests is discussed, highlighting the negative effects on human health, the environment, and the development of pesticide resistance by pests. Manual detection and identification of plant diseases are identified as impractical for large-scale agriculture. Automated methods for crop monitoring and forecasting are needed to reduce pesticide usage, environmental damage, and production costs. This paper acknowledges the potential of big data analysis in the agriculture sector. It highlights the growing importance of machine learning applications in agriculture, as these technologies have proven successful in various domains, including disease detection, image classification, and autonomous vehicles.

The paper presents examples of machine learning applications in agriculture, such as disease classification using Convolutional Neural Networks (CNNs), pest forecasting using Long Short Term Memory (LSTM) networks, and insect detection through deep learning techniques. Commercial tools and services that leverage machine learning are also introduced.

The literature review emphasizes the potential of machine learning to contribute to sustainable agriculture

by assisting farmers in data-driven decision-making, reducing pesticide use, and preserving crop quality and yield. It suggests that ML-based tools can help meet global food demands while minimizing environmental harm. The paper asserts that ML-based techniques have fostered the development of smart farming projects and can contribute to their success by enhancing the capabilities of these projects.

The authors have primarily used Support Vector Machines (SVM), Random Forest (RF), and deep learning-based models for disease and pest classification. SVM and RF attain high accuracy levels, especially in disease classification based on laboratory images, with accuracy of 94%.

The paper [5] emphasize the significance of agriculture in India, highlighting the country's long-standing agricultural traditions and the recent changes driven by globalization. They note that agriculture plays a crucial role in society by providing food, but with a growing population, there is increasing pressure to address food shortages. The authors stress the importance of modernizing agricultural practices through the use of advanced technologies to improve efficiency, profitability, safety, and environmental sustainability. The paper outlines the use of MATLAB for image processing and introduces the Hybrid Neural Network algorithm to enhance the accuracy of soil nutrient and pH level calculations. Furthermore, the paper introduces machine learning, particularly deep learning, as a key component of the proposed system for agricultural yield forecasting.

The authors' novel approach combines deep learning and image processing for soil testing. The system incorporates measurements and observations of soil parameters like moisture, temperature, and humidity, which are then processed using deep learning algorithms like Deep Belief Networks (DBN). The system provides suggestions for optimal crop selection based on these parameters and uses historical water data to assess water availability. The section also introduces the Deep Q-Network (DQN) algorithm, a reinforcement learning approach employed in the system, and explains its architecture and advantages in tackling non-linear relationships and handling continuous observations.

In conclusion, the paper highlights the importance of maintaining soil health and the role of data mining and AI in improving crop productivity. It discusses the potential and limitations of various AI and deep learning techniques in agriculture, especially for soil nutrient prediction and crop yield forecasting. The proposed system represents an innovative approach to addressing these challenges, combining image processing, deep learning, and reinforcement learning to enhance agricultural practices and yield predictions.

The paper [6] introduces a system designed to offer crop and fertilizer recommendations for farmers in India, acknowledging the vital role of agriculture in the country's economy and the availability of extensive historical agricultural data. Information technology is proposed as a solution to process and make recommendations based on this data. Data mining techniques are identified as a means to derive crop and fertilizer suggestions, and the results can be disseminated to farmers through smartphones. The paper focuses on the application of data mining algorithms to build an effective recommendation system by utilizing existing observational data.

The system described in the paper aims to recommend crops suitable for specific regions based on a 3-year crop yield history for that region and suggest appropriate fertilizers for these crops, taking into account soil measurements. These recommendations have the potential to boost crop production for farmers. The random forest algorithm is proposed for crop recommendations due to its efficiency with large datasets and handling of missing values. The paper also explores the use of K-means clustering for predicting the most

suitable fertilizers based on the soil's nutrient content.

In conclusion, the paper presents a system that employs data mining techniques, ontology, and machine learning algorithms to provide crop and fertilizer recommendations to Indian farmers. It aims to increase crop productivity and demonstrates promising accuracy levels in its performance evaluation. Future developments include creating regional language Android applications. This research contributes to the ongoing efforts to leverage technology and data to improve agricultural practices and outcomes in India.

1.4 Research Gap

I identified some research gaps in this domain, after doing a thorough literature survey. The gaps provide us with the opportunity for improvement and innovation in this field. These research gaps are:

- Crop prediction accuracy: The existing systems achieved an accuracy ranging between 90-95%.
- Fertilizer prediction accuracy: The existing systems achieved an accuracy of upto 96%.
- Availability of dataset: Most of the existing systems, are designed using region-specific datasets – for example, designing systems only for the state of Maharashtra or Punjab, etc.
- Existing models either recommend crop or fertilizer, but they have not integrated crop, fertilizer, and pesticide recommendations.
- Most of the systems are developed using very few crops (10-12) and that too region-specific crops.

Chapter 2

Problem Statement

Develop a system that recommends farmers, the right crop to cultivate, based on specific parameters like N-P-K value of soil, pH of the soil, temperature, humidity, and rainfall in the region. Further, the system will also recommend fertilizer and suggestions to farmers based on the crop and N-P-K values and it will also recognize the pests on the crop and recommend pesticide available in India as per ISO standards.

2.1 Research Objectives

The objectives of this project are:

1. Crop Recommendation System: Develop a robust and efficient crop recommendation system that utilizes site-specific parameters such as soil characteristics, soil types, and historical crop yield data to suggest the most suitable crop choices for farmers. The aim is to reduce incorrect crop selection and enhance overall agricultural productivity.
2. Ensemble Model with Majority Voting: Implement an ensemble model with a majority voting technique to improve the accuracy and reliability of the crop recommendation system. This ensemble model should incorporate various algorithms and data sources to provide more precise recommendations.
3. Fertilizer Recommendation: Create a fertilizer recommendation system that takes into account the nutrient content of the soil (N, P, K values) and the selected crop. This system should provide farmers with guidance on the type and quantity of fertilizers required to optimize crop growth and yield.
4. Pest Recognition and Pesticide Recommendation: Develop a pest recognition system capable of identifying common agricultural pests. Integrate this system with a database of pesticides available in India that meet ISO standards. Provide farmers with accurate recommendations for pesticide selection based on the specific pest infestations in their region.
5. Efficiency and Scalability: Ensure that the entire precision agriculture system is designed for high efficiency and scalability. This includes optimizing the computational processes, data collection methods, and user interface to make it accessible and user-friendly for farmers of varying technological backgrounds.

These objectives aim to improve crop selection, enhance agricultural practices, and reduce resource wastage in farming, ultimately increasing productivity and sustainability in the agricultural sector.

2.2 Methodology of the Work

The complete system can be divided into three modules:

1. Crop Recommendation
2. Fertilizer Recommendation
3. Pesticide Recommendation

2.2.1 Crop Recommendation

The steps followed in implementing this module are:

1. Dataset acquisition and preparation:

Collect a dataset that consists of different crops that can be grown under different soil types and weather conditions.

2. ML model Training and creating a .pkl file:

Train the machine learning models on the dataset that was acquired. Different ML algorithms that were used are – SVM, Naïve Bayes, KNN, and Random Forest.

Using the ensemble technique with the majority voting technique, create a .pkl file that considers the ML algorithm, which will give the highest accuracy.

3. Taking User Inputs:

Ask the user to enter appropriate values in the fields mentioned in the UI.

4. Crop Recommendation:

Finally, the system will recommend the crop based on the inputs given by the user, by loading .pkl model file.

ML Algorithms used in building this module are:

i. Support Vector Machine (SVM):

Support Vector Machine (SVM) proves to be a robust machine learning algorithm for crop recommendation system. SVM operates by identifying the optimal hyperplane that distinguishes data points representing different plant species based on their unique features.

SVM is versatile and can handle both linear and nonlinear data classification. When dealing with linearly separable data, SVM seeks the optimal linear hyperplane as a decision boundary that separates one class from another. This hyperplane can be expressed mathematically as:

$W \cdot X + b = 0$, where:

- W is a weight vector ($W = w_1, w_2, \dots, w_n$).
- X is a training tuple.
- b is a scalar.

The optimization problem for the hyperplane boils down to minimizing $\|W\|$.

In cases of linearly inseparable data, SVM employs nonlinear mapping to project data into a higher-dimensional space and subsequently determines a linear hyperplane in this transformed space. Kernel functions play a crucial role in performing these transformations, with options like RBF, linear, sigmoid, and polynomial providing different accuracies.

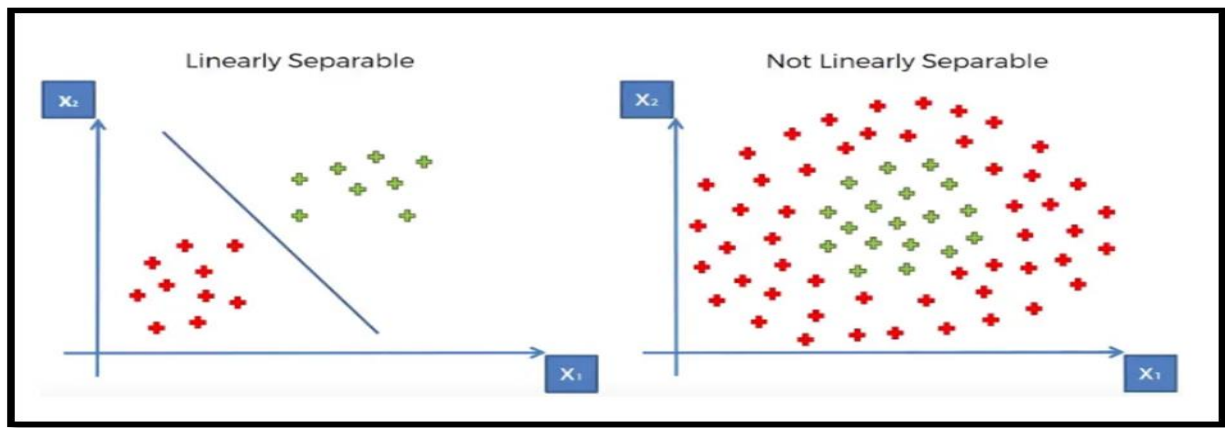


Image 1: Depicting linearly separable and non-separable datasets
https://miro.medium.com/v2/resize:fit:1400/format:webp/1*yAE4_VwfSOk9ohcMtccPg.png

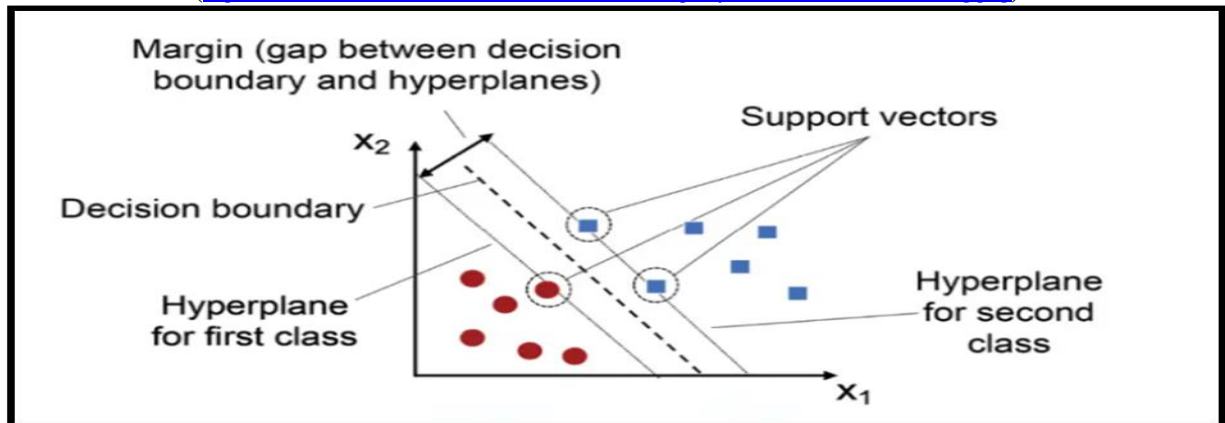


Image 2: Depicting SVM <https://vitalflux.com/wp-content/uploads/2022/08/support-vector-machine-1.png>

The choice of kernel in SVM is pivotal as it transforms data points into a new space, often with a higher dimension, making them more separable. Scikit-learn's SVC algorithm offers various kernel hyperparameter options, including 'linear,' 'poly,' 'rbf,' and 'sigmoid.'

- When using the 'linear' kernel, the model applies a linear boundary, ideal for linearly separable data.
- The 'poly' kernel involves polynomial functions with a degree greater than 1, suitable for complex nonlinear datasets.
- The 'rbf' kernel uses radial basis functions, excelling in handling complex multi-class datasets and noisy data points.
- The 'sigmoid' kernel employs sigmoid functions for classification or regression tasks, with sensitivity to parameter tuning.

Support Vector Machines are powerful tools for classification. However, like any machine learning algorithm, optimizing hyperparameters is crucial for optimal performance. The kernel function is a fundamental hyperparameter dictating data transformation. Regularization parameters such as C and gamma are equally significant, influencing the trade-off between model complexity and overfitting. Other hyperparameters, including the maximum number of iterations, error tolerance, and learning rate, can be fine-tuned to enhance SVM's performance. Through careful adjustment of these hyperparameters, SVM can achieve significantly improved results.

In this project for the SVM algorithm, I used values of degree as degree=1,2,3,4,5 and kernel=poly to generate a model.

ii. K-Nearest Neighbour (KNN):

K-Nearest Neighbour (K-NN) is a simple machine learning algorithm based on the supervised learning technique. The algorithm works by assuming that the similarity between a new data point and the available data points can be used to classify the new data point into a category that is most like the available categories.

In K-NN, all available data is stored, and a new data point is classified based on the similarity to the existing data. This makes it easy to classify new data points into well-suited categories using K-NN. It can be used for both classification and regression problems, but it is mostly used for classification problems. The algorithm is non-parametric, which means that it does not make any assumptions about the underlying data.

K-NN is also referred to as a lazy learner algorithm because it does not learn from the training set immediately. Instead, it stores the entire dataset and performs actions on the dataset at the time of classification.

During the training phase, the K-NN algorithm simply stores the dataset, and when a new data point is added, it classifies the data point into a category that is like the new data. The similarity between data points is measured using distance metrics like Euclidean distance or Manhattan distance.

Overall, K-NN is a simple and effective algorithm for classification problems, which works by finding the K-nearest data points to the new data point and classifying the new data point based on the majority class among the K-nearest neighbours.

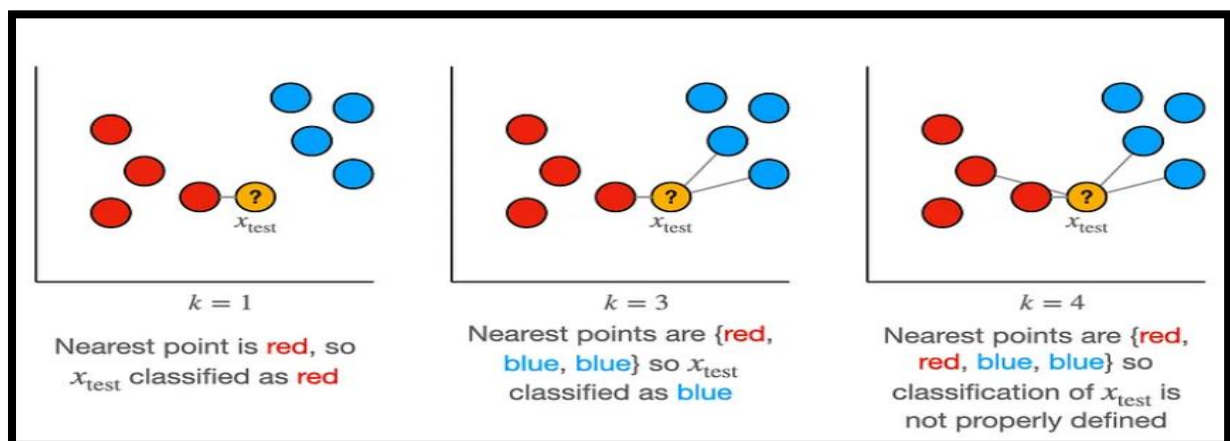


Image 3: Depicting K-NN using different K values (<https://towardsdatascience.com/why-does-increasing-k-decrease-variance-in-knn-9ed6de2f5061>)

The working of the K-NN algorithm can be explained using the following steps:

Step 1: Select the number K of neighbours. K is a hyperparameter that determines the number of neighbours to consider when classifying new data points.

Step 2: Calculate the Euclidean distance between the new data point and each of the K neighbours. The Euclidean distance is a measure of the distance between two points in a multi-dimensional space.

Step 3: Take the K nearest neighbours based on the calculated Euclidean distances. These neighbours are the K data points in the training set that are closest to the new data point.

Step 4: Count the number of data points in each category among the K nearest neighbours.

Step 5: Assign the new data point to the category that has the maximum number of neighbours.

Step 6: The model is ready for classification of new data points.

In this project for K-NN algorithm, I used values of n as n=1,3,5,7,9 to generate a model.

iii. Random Forest:

Random forest is a popular supervised machine learning algorithm used for classification and regression tasks. It works by building multiple decision trees on different samples of the training data and then taking their majority vote (for classification) or average (for regression) to make a final prediction. It is an ensemble method that works by creating multiple decision trees based on different samples of the training data.

To create a random forest classifier, the algorithm takes the training dataset D and creates k bootstrap samples of D, where each sample is denoted as D_i . Each D_i has the same number of tuples as D, but the tuples are sampled randomly with replacement from D. This means that some tuples may occur multiple times in a single D_i , while others may not be included at all. Next, the classifier constructs a decision tree based on each D_i . This results in a "forest" of k decision trees. To classify an unknown tuple, X, each tree in the forest returns its class prediction, and the final decision of X's class is assigned to the class that has the most votes.

The decision tree algorithm used in scikit-learn is CART (Classification and Regression Trees). CART uses the Gini index to measure the impurity of D. The Gini index is computed for each class C_i as the probability p_i that a tuple in D belongs to C_i , and it measures how well D was partitioned. A lower Gini index value indicates a better partitioning of the data.

$$Gini(D) = 1 - \sum_{i=1}^m p_i^2$$

Where P_i is the probability that a tuple in D belongs to class C_i

Thus, the random forest classifier works by creating multiple decision trees based on different samples of the training data, and then combining the results of each tree to make a final prediction. The Gini index is used to evaluate the quality of each split in the decision trees.

In this project for the random forest algorithm, I used 200 estimators that is nothing but a decision tree to generate a model.

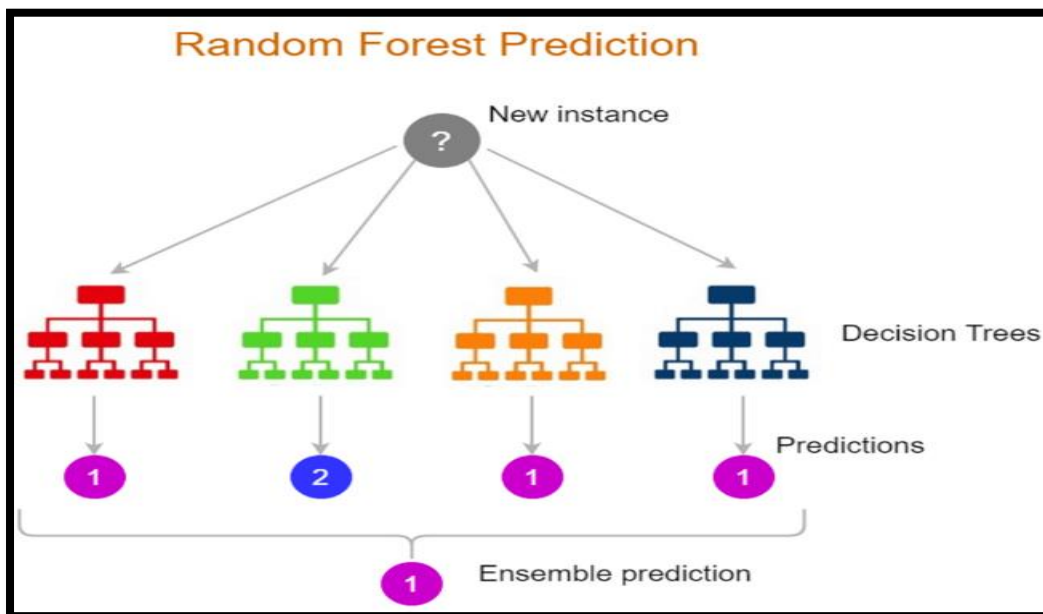


Image 4: Random Forest with 4 Decision trees (https://almablog-media.s3.ap-south-1.amazonaws.com/image_25_7c3fa2cb6f.png)

iv. Naïve Bayes:

Gaussian Naive Bayes is a popular machine learning algorithm used for classification tasks, especially when dealing with continuous or real-valued data. It is a probabilistic algorithm that makes predictions based on the Gaussian (normal) distribution of feature values for each class. It's a part of the broader family of Naive Bayes algorithms, known for their simplicity and efficiency.

Gaussian Naive Bayes is based on Bayes' theorem, which relates the conditional probability of an event (in this case, a data point belonging to a particular class) to the probabilities of the individual features (attributes) of that event. The "naive" assumption in Gaussian Naive Bayes is that all features are conditionally independent, meaning that the value of one feature is not influenced by the values of other features, given the class label.

In the training phase, the algorithm learns the statistical properties of the feature distributions for each class in the dataset. For each class, Gaussian Naive Bayes calculates the mean and variance of each feature, assuming that the feature values follow a Gaussian (normal) distribution. These statistics are used to model the probability distribution of feature values for each class.

When making a prediction for a new data point, the algorithm calculates the likelihood of each class given the observed feature values using Bayes' theorem. For each class, it calculates the product of the probabilities of the features' values for that class (using the Gaussian PDF) and the prior probability of that class. The class with the highest posterior probability (likelihood * prior) is chosen as the predicted class for the data point.

v. Voting Classifier:

A Voting Classifier is a machine learning technique that utilizes an ensemble of multiple models to make predictions. It decides the final output or class based on the highest probability among the options. It accomplishes this by combining the results of each classifier incorporated into the Voting Classifier and making the final prediction according to the majority vote. Instead of developing separate models and assessing their accuracy individually, this approach builds a unified model by learning from these

constituent models and utilizes their combined majority vote for each output class.

Voting Classifier supports two main types of voting:

1. **Hard Voting:** In hard voting, the final predicted output class is determined by the majority of votes from the individual classifiers. For instance, if five classifiers predict the output classes as (A, A, A, A, B), thus the majority predicts class A as the output, so the final prediction will be class A.

2. **Soft Voting:** In soft voting, the output class is predicted based on the average probabilities assigned to each class by the individual classifiers. Consider that three models provide prediction probabilities for class A as (0.70, 0.77, 0.83) and for class B as (0.30, 0.32, 0.40). Calculating the average probability for class A results in 0.76, while for class B, it's 0.34. In this case, class A is the clear winner because it has the highest average probability across all classifiers.

By employing a Voting Classifier, one can harness the strengths of various models and make more robust predictions, thus enhancing the accuracy of machine learning models in various applications.

Now combining all the models that are obtained from the above classifiers Random Forest, Naïve Bayes, KNN, and SVM. In this project, I have used soft-type voting which means the output class is predicted based on the average of probability given to that class.

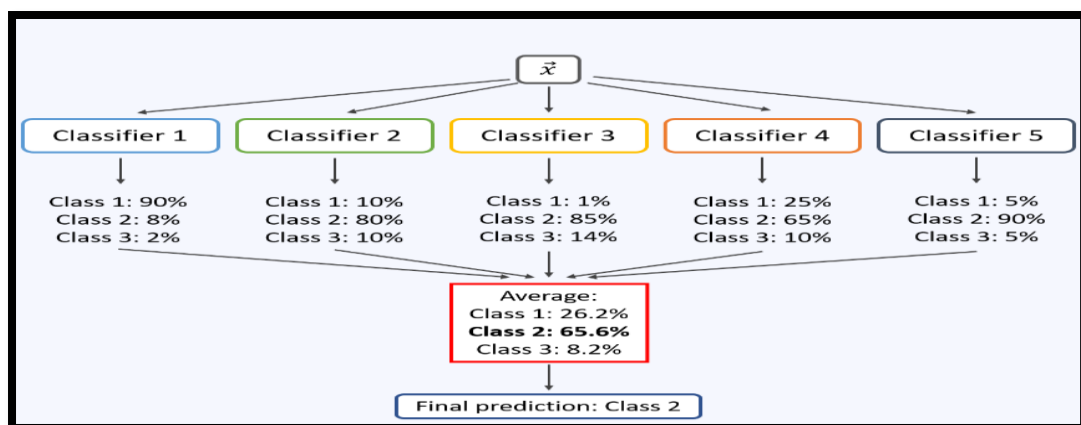


Image 5: Depicting Soft Voting Classifier (https://files.codingninjas.in/article_images/the-voting-classifier-1-1639979194.jpg)

2.2.2 Fertilizer Recommendation

The steps followed in implementing this module are:

1. Dataset Acquisition and preparation:

Collect a dataset that consists of different crops that can be grown under different soil types and weather conditions. The dataset for this project was collected from Kaggle-Build by augmenting datasets of rainfall, climate, and fertilizer data available for India.

2. Taking User Inputs:

Ask the user to enter appropriate values in the fields mentioned in the UI.

3. Calculating the difference between desired and actual NPK values for that crop:

For fertilizer recommendation, the model will check for the level of lack of nutrients in the soil. This can be done simply, by calculating the difference between desired and actual N-P-K values for a particular crop.

Based on the difference calculated, the model will check whether each one of the N, P, and K values are low, high, or present in appropriate amounts in the soil for that crop.

4. Fertilizer Recommendation:

After calculating the difference for each N, P, and K desired and actual values, the model will give dictionary-based suggestions, which are fed to the model by gathering information from various sources. (The Fertilizer Association of India).

2.2.3 Pesticide Recommendation

The steps followed in implementing this module are:

1. Dataset Acquisition and preparation:

Collect a dataset that consists of images of different pests, feeding on crops. For this, I created a dataset by image scraping from Google and then provided corresponding pest labels for all the images.

2. Model training:

This involves training a deep-learning model for pest classification in images. For this project, I have used the CNN model for training the Pesticide Recommendation module. .h5 file is created to store the model.

3. Pest Identification and corresponding Pesticide Recommendation:

.h5 model will be loaded to identify the pest from the given input image and then based on the identified pest, the corresponding pesticide will be recommended.

CNN:

Convolutional Neural Networks (CNNs) are widely used in pattern recognition and image processing as a powerful recognition algorithm. They offer several advantages, such as a simple structure, fewer training parameters, and adaptability.

CNNs are a type of deep learning neural network that has revolutionized image recognition. They are commonly used for visual imagery analysis and are utilized in various applications, from Facebook's photo tagging to self-driving cars, and even in healthcare and security. Image classification involves taking an input, such as an image, and outputting a class or probability that the input belongs to a particular class.

One of the significant advantages of CNNs is that they can automatically extract features from images. They use adjacent pixel information to down-sample the image effectively. A CNN has one or more layers of convolution units that receive input from multiple units from the previous layer, producing a proximity. The input units share or split their weights, and the task of the ConvNet is to reduce the images into a more manageable form without losing the critical features needed for accurate predictions. This is crucial when designing a structure that can learn features and scale to large datasets.

A CNN typically consists of three layers: a convolutional layer, a pooling layer, and a fully connected layer. With these layers, a CNN can effectively process and analyze images to perform various tasks, such as image classification and object detection.

1.Convolution Layer – The Kernel

In a Convolutional Neural Network (CNN), the kernel serves as a filter that extracts features from images. The Convolution Operation aims to identify high-level features like edges from the input image. A CNN

may have multiple Convolutional Layers, with the first layer typically responsible for detecting low-level features such as edges, colours, and gradients. As more layers are added, the network learns to identify high-level features. Convolution is a mathematical operation that combines two sets of information. In CNNs, this operation is applied to input data using a convolution filter, producing a feature map. The Rectified Linear Activation Function (ReLU) is a non-linear piecewise linear function that outputs the input if it's positive; otherwise, it outputs zero.

2. Pooling Layer

The Pooling layer is an important component of convolutional neural networks, which helps to reduce the size of convolved features. This reduction in size is significant, and it plays a crucial role in reducing the computational power required for processing the data. By reducing the dimensions of the data, pooling helps to extract the dominant features, making the training process of the model more efficient.

There are two types of pooling methods that are commonly used in convolutional neural networks.

- (i) Max Pooling- which returns the highest value from the portion of the image covered by the kernel.
- (ii) Average Pooling- which returns the average value of all the values from the portion of the image covered by the kernel.

3. Fully Connected Layer

The Fully Connected Layer (FC Layer) is similar to the Fully Connected Neural Network (FCNN), in that neurons in this layer are connected to all the neurons in the preceding and succeeding layer. It is sometimes referred to as the Dense Layer. By considering all possible combinations of features from the previous layer, this layer learns to map the input representation to the output representation. The flattened output is fed into a feed-forward neural network, and backpropagation is applied during each training iteration. After several epochs of training, the model becomes capable of distinguishing between dominant and low-level features, and the Softmax classification technique is employed to classify them.

Chapter 3

Analysis and Design

3.1 System Configuration

3.1.1 Software Requirements

The software requirements or configuration in this project are:

Operating system- Windows 10/Windows 11/Mac OS/Linux

Libraries- NumPy, Pandas, Sklearn, Keras, OS, OpenCV, Matplotlib, Tensorflow

3.1.2 Hardware Requirements

The hardware requirements in this project are:

Processor- Intel Core i5 or equivalent

RAM- Minimum 4GB or higher

HDD- 256 GB or higher

Architecture- 64-bit

Monitor- 15'' or 17'' display color monitor

Mouse- Scroll or optical mouse

Keyboard- Standard keyboard with 110 keys

3.2 Dataset Details

3.2.1 Dataset for Crop Recommendation

The dataset for this module was collected from Kaggle-Build by augmenting datasets of rainfall, climate, and fertilizer data available for India. (<https://www.kaggle.com/atharvaingle/crop-recommendation-dataset>). Split the dataset into training and testing sets.

Data fields in the dataset are:

- i. N - ratio of Nitrogen content in soil
- ii. P - ratio of Phosphorus content in soil
- iii. K - ratio of Potassium content in soil
- iv. temperature - temperature in degrees Celsius
- v. humidity - relative humidity in %
- vi. ph - ph value of the soil
- vii. rainfall - rainfall in mm

There are a total of 2200 rows in this dataset with 22 different crops.

<i>Index</i>	<i>FeatureName</i>	<i>FeatureDescription</i>
1	Nitrogen (N)	Nitrogen is largely responsible for the growth of leaves on the plant.
2	Phosphorus (P)	Phosphorus is largely responsible for root growth and flower and fruit development.
3	Potassium (K)	Potassium is a nutrient that helps the overall functions of the plant perform correctly.
4	Temperature	temperature in degree Celsius
5	humidity	relative humidity in %
6	pH	pH value of the soil
7	rainfall	rainfall in mm

Image 6: It represents features in the dataset and their meaning (<https://doi.org/10.36227/techrxiv.23504496.v1>)

1	N	P	K	temperatu	humidity	ph	rainfall	label
2	90	42	43	20.87974	82.00274	6.502985	202.9355	rice
3	85	58	41	21.77046	80.31964	7.038096	226.6555	rice
4	60	55	44	23.00446	82.32076	7.840207	263.9642	rice
5	74	35	40	26.4911	80.15836	6.980401	242.864	rice
6	78	42	42	20.13017	81.60487	7.628473	262.7173	rice
7	69	37	42	23.05805	83.37012	7.073454	251.055	rice
8	69	55	38	22.70884	82.63941	5.700806	271.3249	rice
9	94	53	40	20.27774	82.89409	5.718627	241.9742	rice
10	89	54	38	24.51588	83.53522	6.685346	230.4462	rice
11	68	58	38	23.22397	83.03323	6.336254	221.2092	rice
12	91	53	40	26.52724	81.41754	5.386168	264.6149	rice
13	90	46	42	23.97898	81.45062	7.502834	250.0832	rice
14	78	58	44	26.8008	80.88685	5.108682	284.4365	rice
15	93	56	36	24.01498	82.05687	6.984354	185.2773	rice
16	94	50	37	25.66585	80.66385	6.94802	209.587	rice
17	60	48	39	24.28209	80.30026	7.042299	231.0863	rice
18	85	38	41	21.58712	82.78837	6.249051	276.6552	rice
19	91	35	39	23.79392	80.41818	6.97086	206.2612	rice
20	77	38	36	21.86525	80.1923	5.953933	224.555	rice

Image 7: It represents a snap of dataset

3.2.2 Dataset for Fertilizer Recommendation

Data fields in the dataset are:

- N - ratio of Nitrogen content in soil
- P - ratio of Phosphorus content in soil
- K - ratio of Potassium content in soil
- Crop

There are a total of 30 rows in the dataset, representing 30 different crops.

	A	B	C	D
1	Crop	N	P	K
2	apple	20	125	200
3	banana	100	75	50
4	blackgram	40	60	20
5	chickpea	20	50	20
6	coconut	20	10	30
7	coffee	100	20	30
8	cotton	115.7	30.9	7
9	grapes	20	125	200
10	groundnut	35.3	53.8	28.9
11	jute	55.9	22.4	10.2
12	kidneybean	20	60	20
13	lentil	20	40	40
14	maize	120	60	50
15	mango	20	20	30
16	mothbean	20	40	20
17	mungbean	20	40	20
18	muskmelo	100	10	50
19	mustard	81.7	30.4	4.3
20	orange	20	10	10
21	paddy	103.4	32.8	18.8
22	papaya	50	50	50
23	pearlmillet	80	60	40
24	pigeonpea	20	50	20
25	pomegranate	20	10	40
26	rapeseed	81.7	30.4	4.3
27	rice	120	60	60
28	sorghum	58.5	29.1	10.7
29	sugarcane	126.4	45	40.6
30	watermelon	100	10	50
31	wheat	120	60	60

Image 8: It represents a snap of dataset

3.2.3 Dataset for Pesticide Recommendation

The dataset consists of 300 training images per pest and 50 testing images per pest. Total 10 pests are taken into consideration for this project. And for pesticide dataset, it is taken from biostadt website.



Image 9: Pest Dataset - Stem-borer on rice



Image 10: Pest Dataset – Armyworm on maize

3.3 Design

This project supports 22 crops for Crop recommendation: apple, banana, black gram, chickpea, coconut, coffee, cotton, grapes, jute, kidney beans, lentil, maize, mango, moth beans, mungbean, muskmelon, orange, papaya, pigeon peas, pomegranate, rice, watermelon. Hence the user will get results which best suit the land but only from these 22 crops.

The system supports 30 crops for Fertilizer recommendation: apple, banana, black gram, chickpea, coconut, coffee, cotton, grapes, groundnut, jute, kidney beans, lentil, maize, mango, moth beans, mungbean, muskmelon, mustard, orange, paddy, papaya, pearl millet, pigeon pea, pomegranate, rapeseed, rice, sorghum, sugarcane, watermelon, and wheat.

The system supports 10 pests: aphids, armyworms, beetles, bollworms, earthworms, grasshoppers, mites, mosquito, sawfly, and stem borer for the Pesticide recommendation module.

1. In the first module, Crop recommendation, the user is required to input values for N, P, K (all in ratio), temperature (in °C), relative humidity (in %), rainfall (in mm), and pH. Once these values are provided, the system will suggest the most suitable crop for the given land.
2. In the second module, which deals with Fertilizer Recommendation, the user should provide values for N, P, K, and select the crop. Based on this information, the system will offer recommendations for natural fertilizers, taking into account any deficiencies or surpluses of nutrients specific to the selected crop.
3. In the third module, the user should upload a picture that clearly depicts the pest. In this scenario, the system will identify the pest in the background and recommend the corresponding pesticide.

Chapter 4

Results and Discussion

4.1 Results

4.1.1 Result for Crop Recommendation module

Nitrogen (N)

95

Phosphorous (P)

45

Potassium (K)

45

ph level

6.6

Rainfall (in mm)

210

Temperature (in °C)

27

Relative Humidity (in %)

80

Predict

Image 11: Depicting input given for Crop recommendation



Image 12: Output generated for above input

Nitrogen (N)

Phosphorous (P)

Potassium (K)

ph level

Rainfall (in mm)

Temperature (in °C)

Relative Humidity (in %)

Predict

Image 13: Depicting input given for Crop recommendation

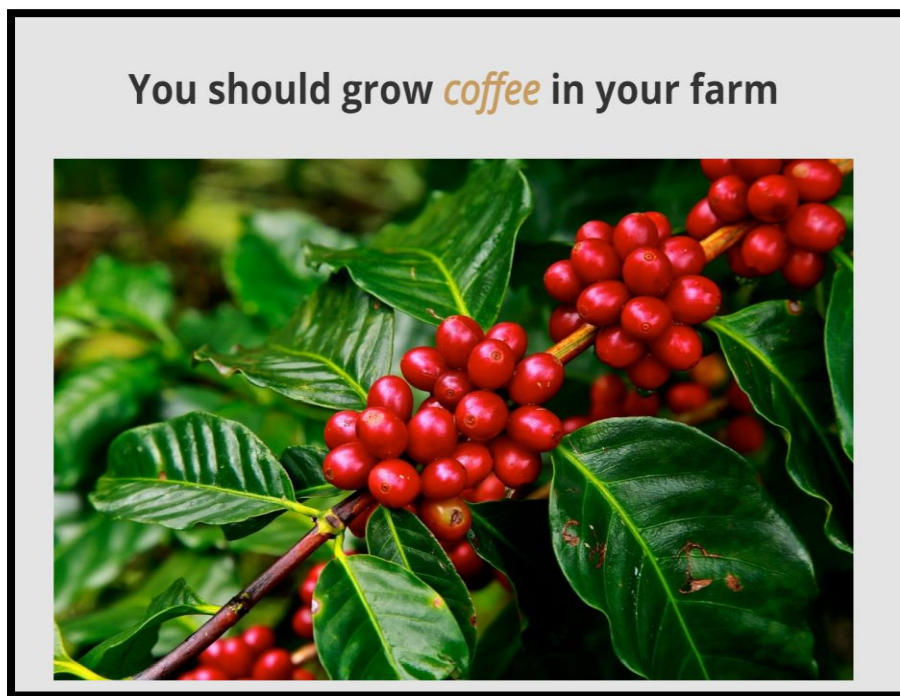


Image 14: Output generated for above input

Nitrogen (N)

Phosphorous (P)

Potassium (K)

ph level

Rainfall (in mm)

Temperature (in °C)

Relative Humidity (in %)

Predict

Image 15: Depicting input given for Crop recommendation



Image 16: Output generated for above input

4.1.2 Result for Fertilizer Recommendation module

Get informed advice on fertilizer based on soil

Nitrogen (N)

80

Phosphorous (P)

40

Potassium (K)

20

Crop you want to grow

lentil

Predict

Image 17: Depicting input given for Fertilizer recommendation

Difference between desired value of N and your farm's N value is 60.0

The N value of soil is high and might give rise to weeds.

Please consider the following suggestions:

1. *Manure* – adding manure is one of the simplest ways to amend your soil with nitrogen. Be careful as there are various types of manures with varying degrees of nitrogen.
2. *Coffee grinds* – use your morning addiction to feed your gardening habit! Coffee grinds are considered a green compost material which is rich in nitrogen. Once the grounds break down, your soil will be fed with delicious, delicious nitrogen. An added benefit to including coffee grounds to your soil is while it will compost, it will also help provide increased drainage to your soil.
3. *Plant nitrogen fixing plants* – planting vegetables that are in Fabaceae family like peas, beans and soybeans have the ability to increase nitrogen in your soil
4. Plant 'green manure' crops like cabbage, corn and broccoli
5. *Use mulch (wet grass) while growing crops* - Mulch can also include sawdust and scrap soft woods

Difference between desired value of P and your farm's P value is 0.0

The P value of your soil is up to the mark.

Difference between desired value of K and your farm's K value is 20.0

The K value of your soil is low.

Please consider the following suggestions:

1. Mix in muricate of potash or sulphate of potash
2. Try kelp meal or seaweed
3. Try Sul-Po-Mag
4. Bury banana peels an inch below the soils surface
5. Use Potash fertilizers since they contain high values potassium

Image 18: Output generated for above input

Get informed advice on fertilizer based on soil

Nitrogen (N)

100

Phosphorous (P)

80

Potassium (K)

60

Crop you want to grow

sugarcane

Predict

Image 19: Depicting input given for Fertilizer recommendation

Difference between desired value of N and your farm's N value is 26.400000000000006

The N value of your soil is low.

Please consider the following suggestions:

1. Add sawdust or fine woodchips to your soil – the carbon in the sawdust/woodchips love nitrogen and will help absorb and soak up and excess nitrogen.
2. Plant heavy nitrogen feeding plants – tomatoes, corn, broccoli, cabbage and spinach are examples of plants that thrive off nitrogen and will suck the nitrogen dry.
3. Water – soaking your soil with water will help leach the nitrogen deeper into your soil, effectively leaving less for your plants to use.
4. Sugar – In limited studies, it was shown that adding sugar to your soil can help potentially reduce the amount of nitrogen in your soil. Sugar is partially composed of carbon, an element which attracts and soaks up the nitrogen in the soil. This is similar concept to adding sawdust/woodchips which are high in carbon content.
5. Add composted manure to the soil.
6. Plant Nitrogen fixing plants like peas or beans.
7. Use NPK fertilizers with high N value.
8. Do nothing – It may seem counter-intuitive, but if you already have plants that are producing lots of foliage, it may be best to let them continue to absorb all the nitrogen to amend the soil for your next crop.

Difference between desired value of P and your farm's P value is 35.0

The P value of your soil is high.

Please consider the following suggestions:

1. Avoid adding manure – manure contains many key nutrients for your soil but typically including high levels of phosphorus. Limiting the addition of manure will help reduce phosphorus being added.
2. Use only phosphorus-free fertilizer – If you can limit the amount of phosphorus added to your soil, you can let the plants use the existing phosphorus while still providing other key nutrients such as Nitrogen and Potassium. Find a fertilizer with numbers such as 10-0-10, where the zero represents no phosphorus.
3. Water your soil – soaking your soil liberally will aid in driving phosphorus out of the soil. This is recommended as a last ditch effort.
4. Plant nitrogen fixing vegetables to increase nitrogen without increasing phosphorus (like beans and peas).
5. Use crop rotations to decrease high phosphorus levels.

Difference between desired value of K and your farm's K value is 19.4

The K value of your soil is high.

Please consider the following suggestions:

1. Loosen the soil deeply with a shovel, and water thoroughly to dissolve water-soluble potassium. Allow the soil to fully dry, and repeat digging and watering the soil two or three more times.
2. Sift through the soil, and remove as many rocks as possible, using a soil sifter. Minerals occurring in rocks such as mica and feldspar slowly release potassium into the soil slowly through weathering.
3. Stop applying potassium-rich commercial fertilizer. Apply only commercial fertilizer that has a '0' in the final number field. Commercial fertilizers use a three number system for measuring levels of nitrogen, phosphorus and potassium. The last number stands for potassium. Another option is to stop using commercial fertilizers all together and to begin using only organic matter to enrich the soil.
4. Mix crushed eggshells, crushed seashells, wood ash or soft rock phosphate to the soil to add calcium. Mix in up to 10 percent of organic compost to help amend and balance the soil.
5. Use NPK fertilizers with low K levels and organic fertilizers since they have low NPK values.
6. Grow a cover crop of legumes that will fix nitrogen in the soil. This practice will meet the soil's needs for nitrogen without increasing phosphorus or potassium.

Image 20: Output generated for above input

4.1.3 Result for Pesticide Recommendation module



Image 21: Depicting aphids pest, input given for Pesticide recommendation

Identified Pest: *aphids*

Recommended Products

		
Dose: 330 ml/acre	Dose: 600 gm/Ha	Dose: 1000 ml/Ha
		
Dose: 30-35 gm/Ha	Dose: 100-150 ml/Ha	Dose: 60-75 ml/Ha

Image 22: Output generated for above input



Image 23: Depicting armyworm pest, input given for Pesticide recommendation

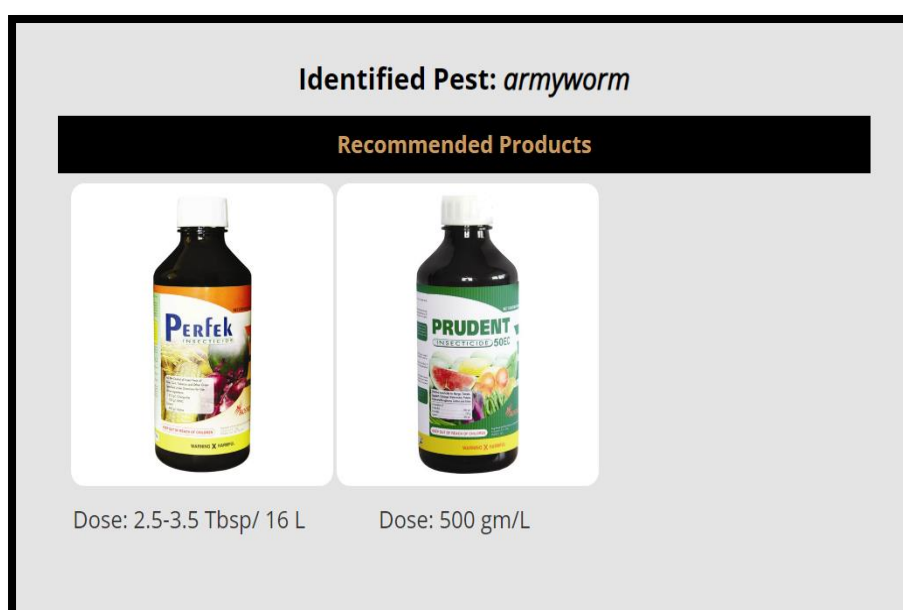


Image 24: Output generated for above input

4.1.4 Accuracy

The average accuracy achieved for the Crop Recommendation system is 98.18%.

The Fertilizer Recommendation system is based on dictionary-based suggestions.

And for Pesticide Recommendation system, CNN achieved an accuracy of 95.2%.

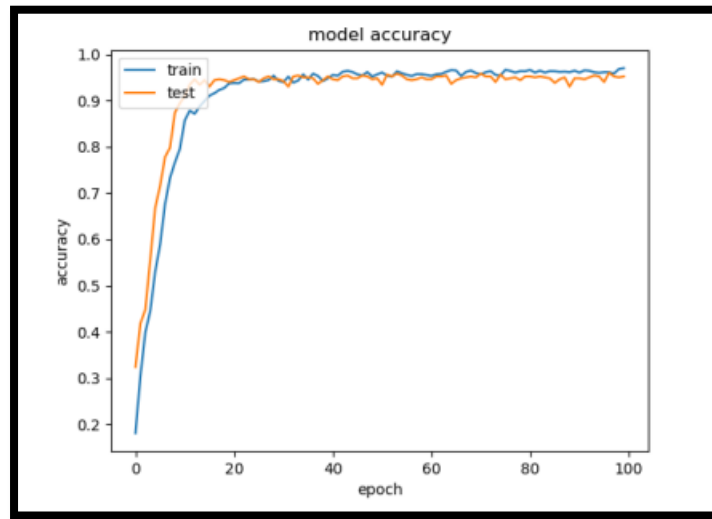


Image 25: Model accuracy vs epochs for CNN

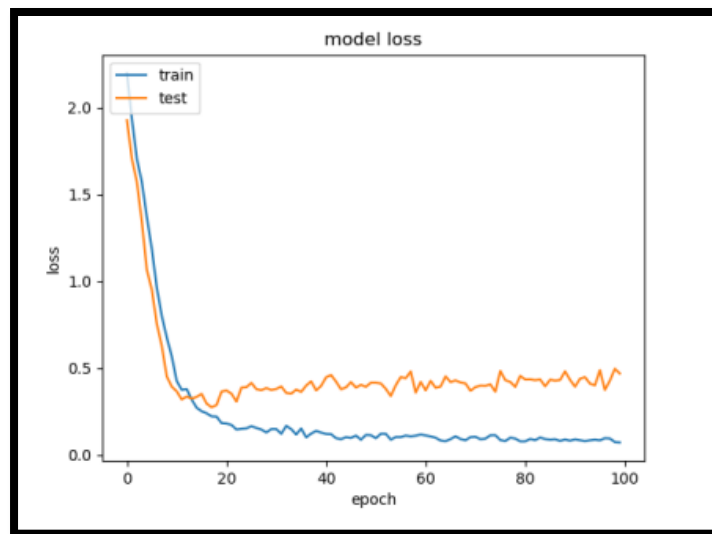


Image 26: Model loss vs epochs for CNN

Chapter 5

Conclusion and Future Scope

Conclusion

India's hardworking farmers play a crucial role in feeding a nation with a population of nearly 1.4 billion people. However, their productivity is constantly threatened by various natural factors that can devastate their crops and, consequently, their livelihoods.

The proposed solution offers a ray of hope for these farmers. It aims to empower them with tools and knowledge to maximize their agricultural productivity. By providing informed recommendations on organic and other fertilizers and assisting in the selection of the right crop based on various attributes, this project offers a comprehensive predictive system that benefits both farmers and the environment. This holistic approach not only enhances crop yields but also ensures the sustainability of agricultural practices.

Moreover, the project also tackles the significant challenge of pest control. Farmers will have access to a reliable system for pest identification and corresponding pesticide recommendations, further safeguarding their crops and livelihoods.

Future Work

This project currently supports 22 crops for the Crop Recommendation system. In future, many other crops can be added to this project, so as to design a robust Crop Recommendation system. Moreover, fertilizers can also be added accordingly.

For the Pesticide Recommendation system, training was done on 10 pests. In future, training can be done on various other pests by gathering their images and more pesticides can be added accordingly.

Some pests, like bollworms, earthworms, and armyworms, look very similar and there are many similar-looking pests. Therefore, in future, better DL models can be used for training, so that our model achieves a higher accuracy.

In future, better UI can be designed, which is more farmer-friendly. UI can be designed in their local languages like Marathi, Hindi, Tamil, Telugu etc.

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